Daily Rant Engineering Design

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1 Introduction

Engineering design is the process of creating solutions to practical problems under real-world constraints. This is done mostly through a Structure Product development Process and Engineering analysis (component sizing) (verfication proofs). The Leonardo Da Vinci's bridge is a portable emergency bridge which functions through three main phenomenon:

- **Friction and Contact:** The crossing point pinch the beam together, so the joint resist slipping
- **Load makes it stronger:** Weight on the bridge presses the beams harder against each other, which tightens the joint.
- **Compression bending:** The structure carries load mostly as compression and bending in the beams, and the geometry spreads that load across many members rather than concentrating in one.

Other factors do affect the design of an Engineering product structurally speaking namely; **material (strain, stress), load, and shape**. It is also important to take into consideration environmental factors and it's durability.

An interdisciplinary project aimed at building an excavator running solely on hydrogen was presented to us in the laboratory. It mainly outputs waster from the chemical reaction shown below:

$$\begin{aligned} \textbf{Anode:} \quad & 2\,H_2 \longrightarrow 4\,H^+ + 4\,e^- \\ \textbf{Cathode:} \quad & O_2 + 4\,H^+ + 4\,e^- \longrightarrow 2\,H_2O \\ \textbf{In Common:} \quad & 2\,H_2 + O_2 \longrightarrow 2\,H_2O \end{aligned}$$

From my understanding electricity is generated through the electrons released during the chemical reaction.

When it comes to Engineering design, we mainly take into consideration the **requirements specification**. This is often gotten from customers and stakeholders by requirements specification specialists, and used for the design of the product. This can further be broken into technical specifications of what is to be designed.

2 Concept

Below is a pictures of the poster for our concept idea for a **Cleaning and Drying station for Wet and Dirty Shoes**. Concept Idea **Smart Carpet** Figure 1.

2.1 Presentation of Concept Idea

The procedure used to come up with this concept was through focusing on requirement rather solely than through solution ideas. This in return created a design that was more realistic and cost efficient. After having looked at a pretty interesting example of a bad design (The rimless bicycle) this oriented my focus on requirements more than an ideology. The main reason why the bicycle idea was a bad design was that, it was solely based on selling the idea of a futuristic looking bicycle rather than something functional. Since using the requirements rout would only produce a regular looking bicycle at the end of the day. In making a good design, we have to take into account the repairability of the product as well as its assembly. This could also be a huge selling point because it makes the usability aspect of the product more intuitive.

2.2 Technical Requirement Specification

- **Size**: Footprint no larger than $90cm$(length) $\times$ $30cm$(width) $= 2.7m^2$ height: $20cm$
- **Capacity**: Capable of cleaning **a pair** of shoe at a time.
- **Material**: Withstand damp shoes, no corrosion, no rot, no rust. The material used should be **recyclable** and **durable**.
- **Cleaning**: Brushes and a Solution of **Water and cleaning Agent**.
- **Humidity**: Keep shoes dry in order to avoid humidity settling in and producing unpleasant smells.
- **Cost**: Cost of materials should be kept under 30 Euros.
- **Low Energy Consumption**: Energy consumption should be kept at the minimum.

There is sunshine for everybody you just have to go outside.

2.3 Description

We are imagining or a product that occupies as much space as a normal shoe rack. Dimensionally speaking we are talking about a rectangular surface area of roughly $2.7m^2$ and a height of $20cm/0.02m^2$. with two compartments, one for cleaning and the other for drying shoes and there is a regular shoe rag for keeping the shoes and letting dry even longer. Refer to the Drawing of the system in the poster Figure 1

2.4 Functioning of the System

The system is divided into two compartments namely; **Cleaning Compartment** and **Drying compartment**

2.4.1 Cleaning Compartment

- The of area has two compartments, where the use is suppose to stand and physically move their fits in a two and fro motion in order to clean each shoe. Inside both compartments we have brushes setup for cleaning the sole of the shoe. Underneath there is the mixture of water and cleaning solution leveled in order to only clean the sole of the shoe(sneakers in this case).

- Water and cleaning agent are stored underneath in what we labeled inside the figure as cleaning compartment. This is suppose to be manually changed over time as the water gets dirty.

2.4.2 Drying Compartment

- The surface of the compartment is similar to that of the cleaning compartment meaning there are two shoe compartments for carrying out the same exact task as the cleaning compartment with the only difference of what is inside the compartment. We have dry reusable towels dry the cleaned shoes.
- Underneath this compartment we also have a bunch of clean dry tissues that automatically gets replaced in a pulling motion to uncover the clean tissues for the next user.
- for shoes that require a quick wipe of the surface, we have included a cleaning wrack on the shoe shelf itself.

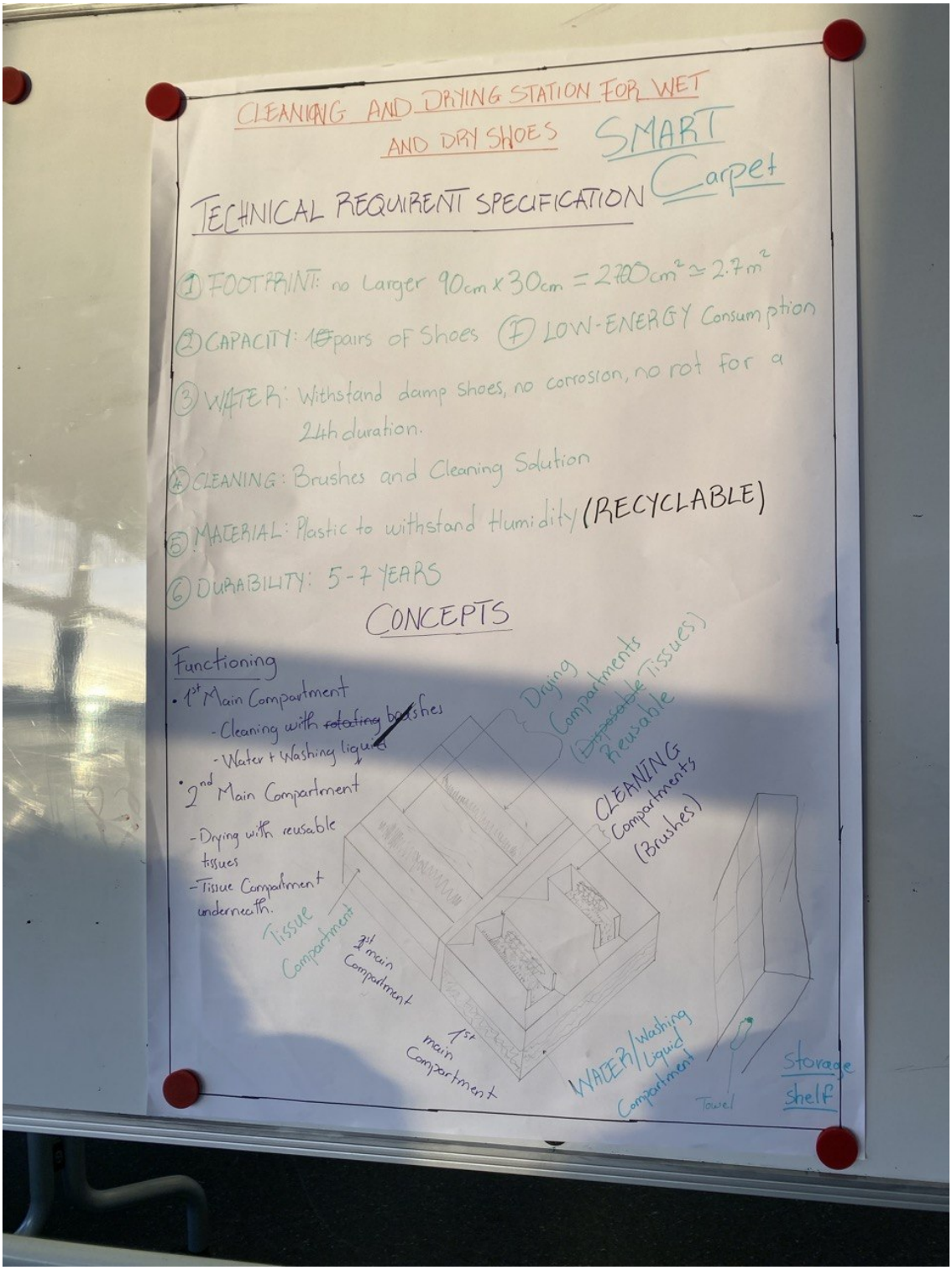


Figure 1: Low cost low energy concept Idea of a cleaning station, where the user simply A rubs their dirty and wet sneakers in the cleaning compartment shown in the figure and dries them on the Drying compartment before taking them off and hanging the in the shoe shelf. The cleaning compartment has a mixture of **water and a cleaning agent underneath**. This water is levelled as per the shoe sole height. The user them subs each shoe on the brushes soaked into the levelled liquid mixture underneath to remove the dirt and moves to the cleaning compartment which contains cleaning re-usable washable cleaning tissues, with a cleaning tissue compartment underneath full of clean replacement tissues. The change from dirty to clean Tissues or towels is down automatically.